

Measuring AI Ability to Complete Long Software Tasks

Time Horizon Doubling Every ~7 Months

Thomas Kwa, Ben West, Joel Becker, et al. — METR | [arXiv:2503.14499](https://arxiv.org/abs/2503.14499) | March 2025

Headline Finding: The 50% task-completion time horizon for frontier AI agents has been doubling approximately every 207 days (~7 months) since 2019, reaching nearly 2 hours with o3.

A New Metric: 50% Task-Completion Time Horizon

A human-grounded, intuitive measure of autonomous AI capability that enables comparison across models from GPT-2 to o3.

1. Assemble Task Suite

170 diverse software / research tasks ranging from 1 second to 8 hours of skilled human effort.

2. Collect Human Baselines & AI Rates

Time skilled humans on each task; measure AI agent success rates under controlled conditions.

3. Fit Logistic Model

Estimate the task duration at which the model achieves exactly 50% success — the time horizon.

Methodology Overview (Figure 2)

Family	Length	Description
find_shell_script	3 seconds	Multiple choice: “Which file is a shell script?” Choices: “run.sh”, “run.txt”, “run.py”, “run.md”
wikipedia_research	1 minute	Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.
oxdna_simple	9 minutes	Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.
munge_data	56 minutes	Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.
cuda_backtesting	8 hours	Speed up a Python backtesting tool for trade executions by implementing custom CUDA kernels while preserving all functionality, aiming for a 30x performance improvement.

170 Tasks Spanning 3 Seconds to 8 Hours of Human Effort

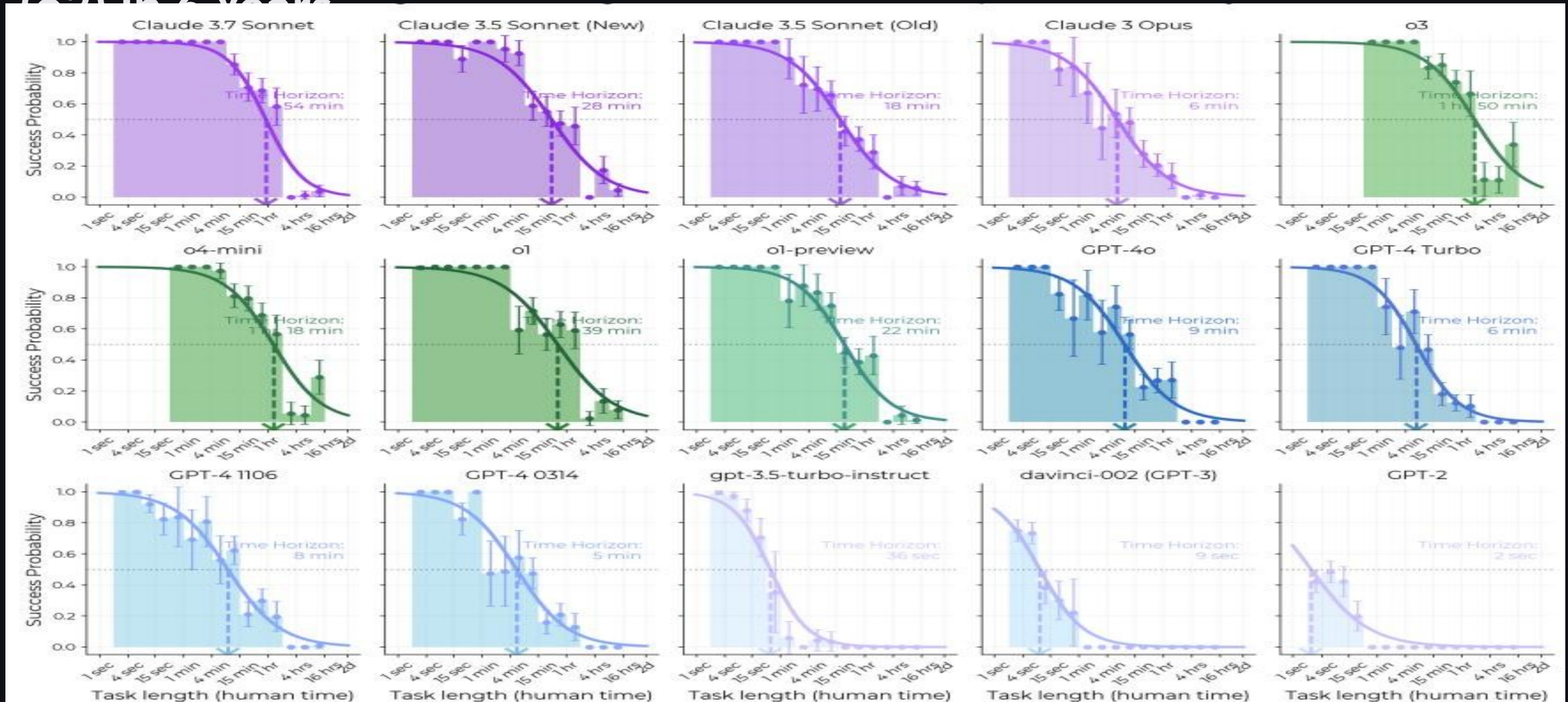
Three complementary benchmarks with >800 human baselines (2,529 hours total).

Source	Tasks	Human Time Range	Description
HCAST	97	1 min – 30 hrs	Diverse software engineering tasks (debugging, refactoring, data munging).
RE-Bench	7	~8 hrs each	Difficult ML research engineering tasks (e.g., custom CUDA kernel backtesting).
SWAA	66	1 sec – 30 sec	Short single-step atomic actions (e.g., identify shell scripts, quick

Example Tasks by Human Duration

Task Family	Human Time	Description
find_shell_script	3 seconds	Multiple choice: 'Which file is a shell script?'
wikipedia_research	1 minute	Research simple factual information from Wikipedia.
oxdna_simple	9 minutes	Detect and fix a bug in molecular dynamics simulation input files.
munge_data	56 minutes	Write a Python script to transform JSON data by inferring conversion rules.

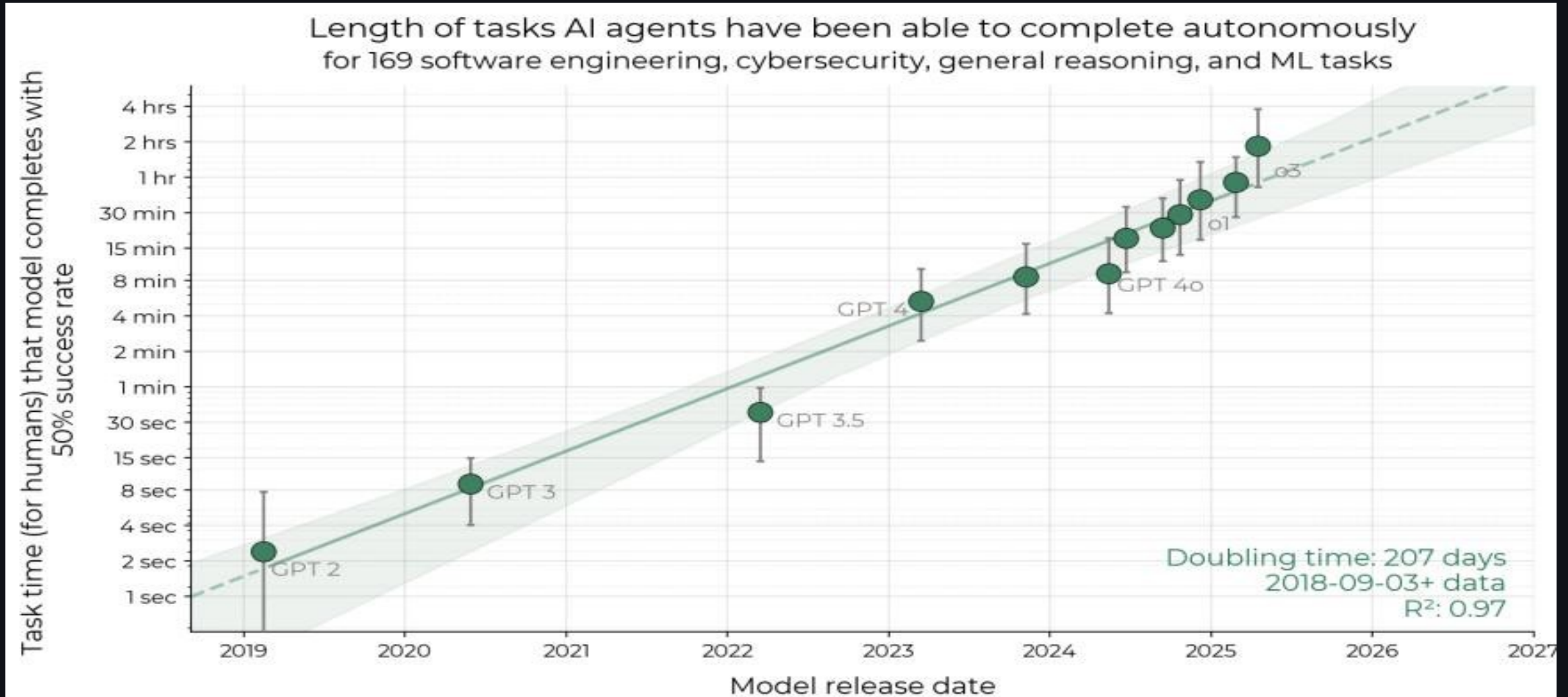
AI Task Horizon Grew from 2 Seconds (GPT-2) to ~1 Hour 50 Minutes (o3) in 6 Years



Doubling time $\approx$ 207 days ($\sim$ 7 months) | $R^2 = 0.97$ | 2023-2025 growth $\sim$ 20% faster than 2019-2025 baseline

Success Probability Drops Sharply as Task Length Increases for All Models

Frontier models push the curve rightward: Claude 3.7 Sonnet (~54 min) and o3 (~1h 50m) extend the tail far beyond GPT-2 (~2 sec).



The 80% time horizon follows a similar trajectory but remains roughly 5× shorter than the 50% horizon.

Reasoning, Tool Use, and Error Recovery Drive Time Horizon Growth

- Improved logical reasoning — models plan and decompose multi-step problems more effectively.
- Better tool use — integration with IDEs, terminals, and APIs expands actionable scope.
- Greater reliability and self-awareness — models recognize when they are stuck and backtrack.
- Ability to adapt to mistakes — error recovery reduces cascading failure on long tasks.

Key Limitations

- Performance is notably lower on less structured, 'messier' tasks that lack clear success criteria.
- External validity remains open — benchmark tasks may not perfectly represent real-world intellectual labor distributions.
- Supplementary experiments found little evidence that trends are slower on more realistic tasks, but generalization is not guaranteed.

Extrapolation Predicts Month-Long Task Automation by 2028–2031

If the exponential trend continues, frontier AI would autonomously complete tasks requiring ~167 work hours (1 month) within the next 3–6 years.

2019	2023	2024	2025	2028–2031
GPT-2	GPT-4	o1 / Claude 3.5	o3 / Claude 3.7	Projected frontier
~2 seconds	~5 minutes	~28–39 minutes	~54 min – 1h 50m	~1 month (167 hrs)

Caveats

- Extrapolation assumes no fundamental slowdown or saturation in scaling laws.
- External validity: month-long benchmark tasks may still differ from messy, real-world software projects.
- 2023–2025 growth rate is ~20% faster than the 2019–2025 average, suggesting possible acceleration — or a transient burst.

Takeaways: Exponential Autonomous Capability Growth Demands Proactive Safety Governance

1. Robust metric — The 50% time horizon is intuitive, human-grounded, and tracks autonomy with $R^2 = 0.97$ over 6 years of model releases.
2. Consistent doubling — The horizon doubles every ~7 months; post-2024 data hints at acceleration, warranting close monitoring.
3. Current frontier — o3 achieves ~1 hour 50 minutes of human-equivalent autonomous work, up from 2 seconds for GPT-2 in 2019.
4. External validity is an open question — tasks are structured; performance on unstructured, 'messy' real-world labor remains uncertain.
5. Governance imperative — Month-long automation by 2028–2031 would transform software R&D and labor markets; safety protocols, eval regimes, and policy frameworks must keep pace with this exponential trajectory.